



DA VINCI'S CIVILS & PUMPS MANAGEMENT APPLICATIONS

Proposal and Quotation

Date of issue: 29 January 2025

Compiled by MKat





Table of Contents

Scope of Work	4
Features of the Application	4
DVCP Client Interface.....	4
DVCP Admin Interface	8
DVCP Technician App.....	10
Timeline and Milestones.....	15
Cost Breakdown	18
Setup Fees	18
Service Fees and Expansion Procedures.....	20
Service Level Agreement (SLA)	23
Billing Terms.....	23
Additional Notes.....	23
Billing & Reconciliation.....	23
Standard payment terms.....	23
Milestone requirements	23
Change Request Process	24
Document Digitisation	24
Initial Period (First 3 Months).....	24
Payment Terms	24
Documentation Requirements.....	24
Approval Workflow	25



List of Figures

Figure 1 - Login	5
Figure 2 - Alternate Contact options	5
Figure 3 - Client Dashboard	6
Figure 4 - New Service Request	6
Figure 5 - Active Service Requests	7
Figure 6 - Client Reporting	7
Figure 7 - Client Equipment Management	7
Figure 8 - Client Services Menu	7
Figure 9 - Unprocessed Requests	8
Figure 10 - Ongoing Jobs	9
Figure 11 - Technician Login	11
Figure 12 - Technician Job Card	11
Figure 13 - Proof of Safety Compliance	11
Figure 14 - Documentation Completion	11
Figure 15 - Document Parts Installed	12
Figure 16 - Technician Job Completion	12
Figure 17 - Job Card Equipment Details	13
Figure 18 - Job Card Checks	13
Figure 19 - Job Card Readings	14



Scope of Work

This document describes MKat's proposed solution for Da Vinci's Civils & Pumps (DVCP). This solution will allow DVCP to **log client job requests**, **issue job cards**, and **oversee job completion**. Also described is the potential integration into a Petroleum Safety Training Application (PSTA) that allows individuals to undergo training to acquire safety badges that can be required to be assigned to a job card. The PSTA is not included in this proposal and quotation but will be referenced for how the DVCP solution may integrate and leverage its capabilities to expand its capabilities.

Features of the Application

The solution described here (subject to alterations requested by the client) has three main user types, each of which will have a tailored experience in the application based on their roles. The following sections will group the proposal by the interface pages, which the specific users will interact with.

The three users are:

- Admin
- Technicians
- Clients

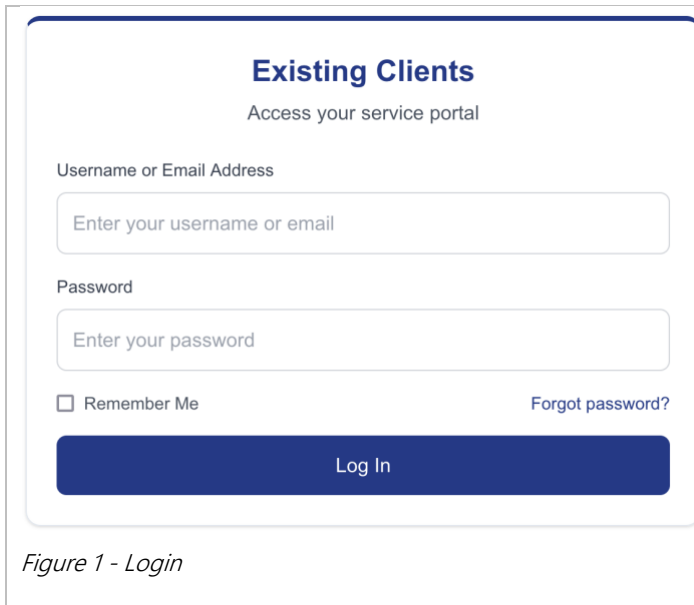
The following subsections will describe these users and the pages they interact with.

DVCP Client Interface

An app that allows the DVCP clients to log an enquiry for services to be rendered. This will be available from the website.

This is where any new job starts. When a client (Petrol Station) needs to request DVCP's services, they have a range of contact options; they can either WhatsApp, email or phone the office or use the website to capture the job request. In the event that the client chooses not to use the website, the DVCP staff will act on their behalf by capturing the request on the system.

A demonstration of this application can be found at: <http://102.37.23.105:3005/>



Existing Clients
Access your service portal

Username or Email Address

Enter your username or email

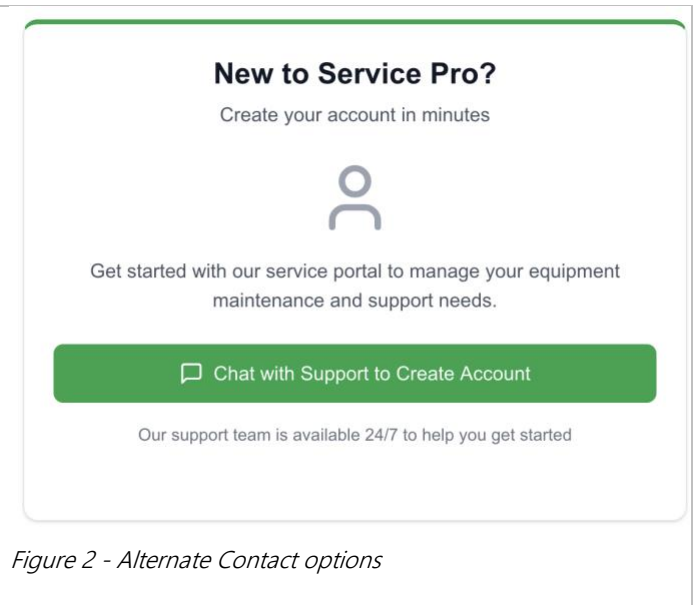
Password

Enter your password

☐ Remember Me [Forgot password?](#)

Log In

Figure 1 - Login



New to Service Pro?
Create your account in minutes



Get started with our service portal to manage your equipment maintenance and support needs.

Chat with Support to Create Account

Our support team is available 24/7 to help you get started

Figure 2 - Alternate Contact options

Once a user logs in (Figure 1), they will be greeted by their dashboard (Figure 3) where they can create a new service request. The creation of a new service request prompts the client for information pertaining to the job (Figure 4). These include:

- The location of the site
- The ID or serial number of the equipment that requires service (as well as visual evidence)
- A description of the issue
- The priority level required for the response.

Should the client be found to have inflated their priority level, DVCP reserves the right to charge a penalty fee and reduce the priority. This functionality will be described the Admin section.

Once the service request has been created it will be added to the list of active service requests which the user can also view from their dashboard (Figure 5). A job can then be created and the admin team can create the job card and dispatch a maintenance team accordingly. This list will also depict the status and progress of the service requests as well as the technician who has been assigned to the job.

Jobs can be summarised by a progress report that the owners can download (including how many hours have been completed). The financial threshold must also be depicted here, this is a progress bar that conveys how much off the allocated threshold budget has been used, when the threshold is met, all work is ceased until the client agrees to continue.

In addition to creating new service requests, clients can also review their history of completed jobs (Figure 6). The equipment on site can also be monitored (Figure 7).

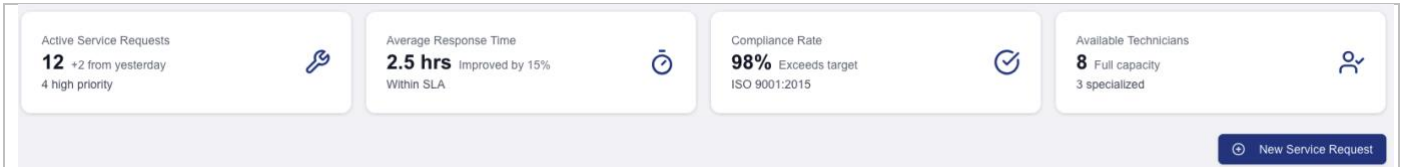
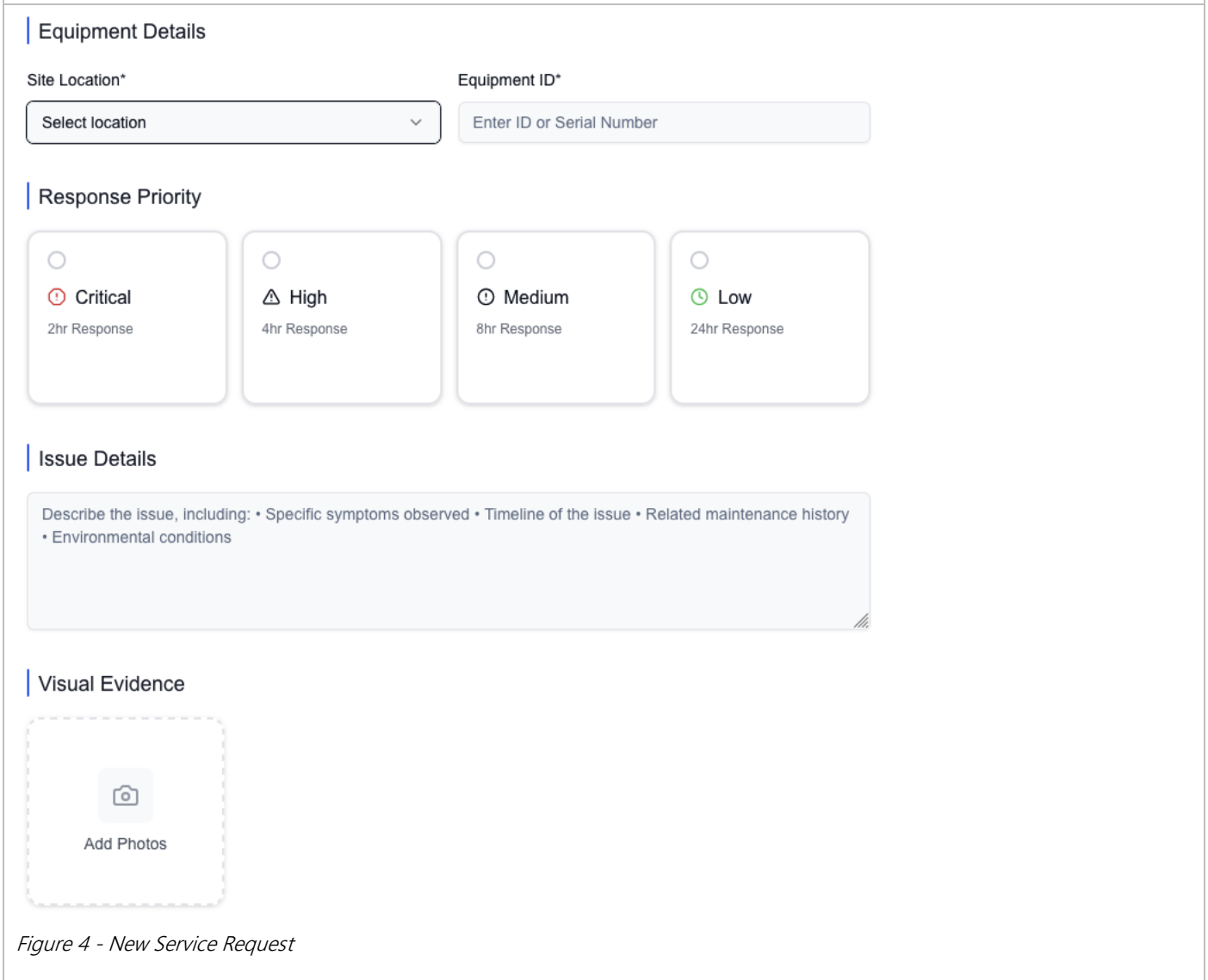


Figure 3 - Client Dashboard



Equipment Details

Site Location* Equipment ID*

Response Priority

☐ **Critical**
2hr Response

☐ **High**
4hr Response

☐ **Medium**
8hr Response

☒ **Low**
24hr Response

Issue Details

Describe the issue, including: • Specific symptoms observed • Timeline of the issue • Related maintenance history • Environmental conditions

Visual Evidence

Figure 4 - New Service Request



Active Service Requests

Real-time status of ongoing service requests

Q Search by ID, site,...

All Statuses

All Priorities

Reference	Site	Equipment	Issue	Priority	Status	Progress	Technician	ETA	Actions
SR-2024-001	Station Alpha	Fuel Dispenser	Flow Rate Deviation	High	In Progress	65%	John Smith	2 hours	:
SR-2024-002	Beta Terminal	Storage Tank	Level Sensor Calibration	Medium	Assigned	20%	Sarah Johnson	4 hours	:
SR-2024-003	Gamma Station	POS Terminal	Payment Processing Error	Critical	Overdue	0%	Unassigned	Overdue by 48 hours	:
SR-2024-005	Epsilon Fuel Stop	Vapor Recovery	Compliance Check	High	Overdue	15%	David Chen	Overdue by 24 hours	:
SR-2024-006	Zeta Highway Stop	Emergency Generator	Scheduled Maintenance	Medium	Scheduled	0%	Lisa Martinez	Next Week	:
SR-2024-007	Eta Express	Price Sign	Display Malfunction	Medium	In Progress	45%	Tom Brown	3 hours	:
SR-2024-008	Theta Services	Underground Piping	Pressure Test Required	Critical	On Hold	30%	James Wilson	Pending Parts	:
SR-2024-009	Iota Junction	Car Wash Equipment	Water Pressure Low	High	Pending Customer	75%	Alex Thompson	Awaiting Approval	:

Figure 5 - Active Service Requests

Reports

Service History

Analytics

Compliance Docs

Safety Records

Equipment Status

Fuel Systems

Pumps & Dispensers

Tank Monitoring

Performance

Service Center

Active Tickets5

Scheduled Service

Calibration

Compliance

Figure 6 - Client Reporting

Figure 7 - Client Equipment Management

Figure 8 - Client Services Menu



DVCP Admin Interface

An app that allows the DVCP administrative staff to manage jobs on sites.

Once a service request is created by the user, or by the DVCP staff on behalf of the client, it will appear on the Admin Interface until it is processed (Figure 9). The client requests will be handled manually by the DVCP staff. Once They have handled the requests they use their interface to create job cards and assign them to technician teams (Figure 10).

A demonstration of this application can be found at: <http://102.37.23.105:3006/>

Unprocessed Requests

Ongoing Jobs

Search by ID, site...

Create New Request

Unprocessed Service Requests							
Service requests awaiting processing and assignment							
Reference	Site	Equipment	Issue	Priority	Technician	ETA	Actions
SR-2024-001	Sandton Depot	Fuel Dispenser	Flow Rate Deviation	High	Unassigned	Pending Processing	⋮
SR-2024-002	Durban Terminal	Storage Tank	Level Sensor Calibration	Medium	Unassigned	Pending Processing	⋮
SR-2024-003	Pretoria Station	POS Terminal	Payment Processing Error	Critical	Unassigned	Pending Processing	⋮

Figure 9 - Unprocessed Requests



Unprocessed Requests

Ongoing Jobs

Search by ID, site...

Create New Request

Ongoing Job Cards

Currently active service requests being processed

Reference	Site	Equipment	Issue	Priority	Technician	ETA	Actions
JC-2024-001	Cape Town Terminal	Vapor Recovery	Compliance Check	High	David Naidoo	2 hours	
JC-2024-002	Bloemfontein Stop	Emergency Generator	Scheduled Maintenance	Medium	Sarah van der Merwe	Next Week	

Figure 10 - Ongoing Jobs

As maintenance is performed, a detailed database will be built for each site, eventually establishing a record of every part on every site to allow easy search and glean which parts are problematic where.

DVCP Technician App

An app that allows the DVCP technicians to log jobs and fulfil the required legislative and safety requirements that are involved in a Job

Once the admin team processes a service request and uses their interface to create and assign technicians to a job card, the technicians will be notified via SMS, email, WhatsApp and the DVCP Technician App.

This app will require the technicians to log in (Figure 11) and enter a job card number (Figure 12).

For a new job, all of the required documentation will be available for the technician to present to the client for signature (Figure 14).

Once the documentation has been completed, the app will prompt the technician for photographic evidence of the Personal Protective Equipment required to complete the job (Figure 13). Once all of these steps have been completed, the technician will be allowed to access additional menus that allow them to complete the job.

All parts which are installed or serviced have a brand and serial number which must be captured in the app along with photo evidence of both the new and old parts (Figure 15). Any maintenance or procedural documentation which cannot be digitised must be uploaded photographically as well. Furthermore, in the event of an accident, technicians will be able to compile and submit incident reports. These will help improve both training and the sites serviced.

The above can all be automated into a WhatsApp bot that automatically requests and captures the above information for the technician for ease of use.

Furthermore jobs that require more detailed forms can be assigned more detailed job cards as seen in Figure 17. All checks must be performed before work can commence (Figure 18) and the details of the job can be entered (Figure 19). These job cards are digitised for ease of use in the application and can be exported into a PDF format for documentation purposes. Should the document need to take a special format for legal reasons, a PDF viewer can host these forms in the application.

A demonstration of this application can be found at: <http://102.37.23.105:3002/>

Login

Enter your credentials to access your assigned jobs



Login

 Report Incident

Figure 11 - Technician Login

Job Card Entry

Enter or scan your job card number

Enter Manually

Scan QR Code

 Report Incident

Figure 12 - Technician Job Card

Safety Requirements

Complete safety documentation and PPE verification


PPE Verification

Safety Helmet

 Upload

Safety Boots

 Upload

Safety Gloves

 Upload

Confirm Safety Checks

 Report Incident

Figure 13 - Proof of Safety Compliance

Client Documentation

Obtain necessary client signatures and approvals


Required Signatures

Work Authorization Form

Safety Compliance Form

Continue to Work

 Report Incident

Figure 14 - Documentation Completion

Work Execution

Document parts and complete required procedures

Parts Documentation

Old Part

New Part

Additional Documentation

Figure 15 - Document Parts Installed

Job Completion

Review and submit completed job

Job Complete

All required documentation has been collected and verified.

Job Summary

Job Number:

Safety Checks:

Client Signature:

Parts Logged:

Completed

Obtained

Verified

Figure 16 - Technician Job Completion

Equipment Details

Enter equipment and issue information

Equipment Information

Make & Model (e.g., Dresser Wayne Helix)

Serial Number

Grade (e.g., ulp93 and dsl50)

Contact Number

Issue Description

Continue to Safety Checks

 Report Incident

Figure 17 - Job Card Equipment Details

Safety Checklist

Complete all required safety checks before proceeding

Dealer's consent to commence work?

☐ Y ☒ N

Protective clothing and safety shoes are being worn?

☐ Y ☒ N

Barriers erected around the work place?

☐ Y ☒ N

Fire extinguisher (9kg) accessible within the work place?

☐ Y ☒ N

Electrical supply isolated and tagged?

☐ Y ☒ N

Fuel supply to the pump or dispenser isolated?

☐ Y ☒ N

Continue to Work

 Report Incident

Figure 18 - Job Card Checks

Work Execution

Document work carried out and parts used

Work Details

Describe work carried out

Root Failure Cause

Not Pump Related

Meter Readings

Litre Tote Before

Litre Tote After

Parts Used

S/Code

Description

Add Another Part

Faulty Items Returned

S/Code

Description

Add Another Item

Complete Job

Report Incident

Figure 19 - Job Card Readings

Timeline and Milestones

Following the last meeting, the application timeline has been revised to allow for rapid deployment; those revisions are detailed here. The application will be deployed in **2 stages**.

1. **Stage 1:** Fully functioning apps, with limited capabilities in certain interfaces to allow for a rapid launch.
2. **Stage 2:** Fully functioning apps with full feature set described earlier.

* Please note that Stage 1 will feature the full feature set for both the client and admin portals. The only app whose function will be limited in this stage will be a simplified technician app. The technician app in this stage will still notify technicians of new jobs and provide them with instructions but will not feature digitised forms and job cards. In this stage, the app will allow these documents to be scanned in for storage and processing.

The stages will be delivered in a phased approach (**Error! Reference source not found.**). Each of the above phases has an associated milestone (Table 2). Milestones are reached upon a client review at which time the following phase commences. The phase durations listed above each start upon the completion of the preceding milestone.

* Note that the Design Approval phase can be accelerated should the client be satisfied with the application presented earlier in the document and only require minor revisions.

Table 1 - Proposed Phases for Project Completion

PHASE		DURATION (WEEKS)
STAGE 1		
1.1	Design Approval	0
1.2	Development Completion	3
1.3	Testing	1
1.4	User Acceptance Testing	Ongoing
STAGE 2		
1.1	Design Approval	2
1.2	Development Completion	6
1.3	Testing	2
1.4	User Acceptance Testing	3
1.5	Final Deployment Initial Support (at no additional charge*)	(30)



Total Time to Deployment + User Acceptance Testing each phase is subject to a buffer time of 3 weeks	10 + 3
--	---------------

* The initial support period allows for any customer feature requests at no additional charge provided the features incur no additional costs with MKat's cloud partners.

Table 2 - Phase Milestones and Breakdown.

PHASE		Milestones and Breakdown
1	Design Approval	Activities: 1. Prototype Application Design Development 2. Prototype Application Design presented to Stakeholders (meeting) * 3. Revisions to App design based on Feedback * 4. Final Application design presented to Stakeholders Milestone: <i>Client Sign-off on Proposed Application Design</i>
2	Development Completion	Activities: 1. Prototype Application Development 2. Migrate existing documents and Data 3. Prototype Application presented to Stakeholder (meeting) 4. Revisions to App, based on Feedback * 5. Final Application design resented to Stakeholders Milestone: <i>Client Sign-off on Developed Application</i>
3	Testing	Activities: 1. Internal Application testing of the different user interfaces 2. User documentation created. Milestone: <i>mkat Sign-off on Application</i>
4	User Acceptance Testing	Activities: 1. Onboarding and training of the various user types in application use 2. Documentation for Application use is made available 3. Training of the various user types in Bug report compilation 4. Application fixes for reported bugs 5. Application fixes presented to Stakeholders Milestone: <i>Client Sign-off on Application Function after Testing</i>
5	Final Deployment	Activities: 1. Application is Launched as is available to all Stakeholder user types 2. Ongoing support for changes and additions to the apps from mkat for 6 months at no additional cost (provided the changes remain within the chosen user and server tier)

* Multiple feedback sessions can be conducted with stakeholders to refine the application to their needs.



Cost Breakdown

The cost is broken into two sections: Setup and Service Fees. Setup Fees cover the migration of data and relationships required for the use of the Application as well as its hosting during the user-testing phase. Service Fees cover the hosting of the application and its associated data created through using the application following the completion of setup. Service fees are only required once the setup phase is complete (marked by: *Client Sign-off on Application Function after Testing*)

A Rebate is offered on both setup and service fees if the client can guarantee long-term usage of the application. As such two rebate tiers are offered, a 24 and 36-month plan. Longer periods equate to more substantial rebates.

Following the last meeting and the need for rapid initial deployment, the pricing is revised and documented by stage. These stages will be described in the relevant sections.

Setup Fees

Setup fees cover the process of converting the customers' data and workflow into the application as well as tailoring the application's framework to suit the customers' particular needs.

Following the satisfaction of the customer, documentation will be signed to mark the completion of each phase and the fulfilment of the associated milestone. Select milestones will be followed by a payment for the completion of the associated setup. A breakdown of the payments for these milestones are presented in Table 3. The fees contained in this table are divided into the stages and their associated phases described earlier.



Table 3 - Development Cost Breakdown (All totals include 15% VAT)

		PAYMENT DUE (R) FOR CHOSEN PLAN		
STAGE 1				
PHASE	Milestone	Standard	24 Month Rebate	36 Month Rebate
1.1	Client Sign-off on Proposed Application Design	R246,718.97	R209,711.13	R155,432.95
1.2	Client Sign-off on Developed Application	R411,198.29	R349,518.54	R259,054.92
1.4	Client Sign-off on Application Function after Testing	R164,479.31	R139,807.42	R103,621.97
STAGE 2				
2.1	Client Sign-off on Proposed Application Design	R164,479.31	R139,807.42	R103,621.97
2.2	Client Sign-off on Developed Application	R274,132.19	R233,012.36	R172,703.28
2.4	Client Sign-off on Application Function after Testing	R109,652.88	R93,204.94	R69,081.31
	Total	R1,370,660.95	R1,165,061.81	R863,516.40



Service Fees and Expansion Procedures

This section details the monthly service fee due to MKat for the ongoing hosting and licensing of the application. Pricing is tiered based on the number of application users. Rebates are available for higher user tiers as well as long-term subscriptions. Service Fees begin following the completion of the application setup (marked by the signing of: *Client Sign-off on Application Function after Testing*)

A tier must be selected for the hosting of the application. The choice of the tier determines the performance and scope of the application. Tiers can always be upgraded should the usage of the application outgrow the current tier.

The offerings and pricing of the different tiers are described in Table 4.

The user groups who interact with the application, the quantities of whom determine the tier, are described in Table 5. This table can be used as a guide to the tier required by the customer. Likewise, the servers used to host the application and their attributes for the various tiers are discussed in Table 6.

Increased numbers of users increases the demand on the server and the scale of the data that needs to be stored. As such the Service Fee may need to be increased when the user count exceeds the limits or when the volume of data being used per user exceeds the limits of the current tier.

MKat provides a post-launch support period of 6 months from the completion of the testing period (after the signing of the *Client Sign-off on Application Function after Testing*). During this post-launch period, new features and alterations can be requested and added at ***no additional cost***, provided that these features do not require a change in user or server tier and do not incur additional costs with MKat's cloud partners.

MKat has a partnership with Microsoft Azure to host the application and the associated data on the Microsoft Servers in Gauteng. This hosting choice ensures that the customer's data is not only securely stored on world-class servers, but that the running of the application cannot be compromised by any breaks in underwater cables or other external factors that would negatively impact applications hosted on foreign servers.

Table 4 - User Tier Offerings

Tier	Capturers	Coordina	System	Sites	Server	Standard Fee	High-User Count Rebated Fee (complimentary)	24 Month Rebated Fee	36 Month Rebated Fee
Platinum	1000	50	20	500	Enterprise	R149,376.95	R131,451.72	R124,879.13	R118,306.55
Gold	500	25	10	250	Professional	R80,784.14	R76,744.93	R72,907.69	R69,070.44
Silver	200	10	5	100	Professional	R61,064.52	R59,232.59	R56,270.96	R53,309.33
Bronze	100	5	1	50	Basic	R31,427.53	R31,427.53	R29,856.16	R28,284.78
Bronze Lite	100	5	1	50	Basic	R21,359.74	R21,359.74	R21,359.74	R21,359.74

* Note that the High-User Count Rebate is complementarily included for all tiers above the Bronze tier.

* Note that Bronze Light is only available for Stage 1 of the deployment, with a maximum period of 4 months. Service Fees will automatically upgrade to Bronze following the completion of Stage 2 (Client Sign-off on Application Function after Testing).

Table 5 - User Type Descriptions

User Type	Permissions	Description	DVCP User Type
Capturers	Log information. Access to only their own data	Technicians and Site owners who log and create service requests	- Technicians - Clients (Service Station Owners/Managers)
Coordinators	Process Information and create Jobs for Capturers. Access to all capturer information	Admin, planning and HR who process service requests and create job cards	Administrative Staff
System Admin	Full access to all data and ability to Create and Delete users and sites	These users manage how all other users use the application	Senior Staff
Sites	n/a	All jobs and replacement parts will be stored relative to the site where they were performed	Service Stations

Table 6 - Server Offerings

Server	Description
Enterprise	For Large Scale Operations with thousands of daily users
Professional	Improved speeds for concurrent users, allowing for many users to be interacting with the server at once with no latency.
Basic	Entry level server, limited speeds for many concurrent users.



Service Level Agreement (SLA)

1. 96% uptime guarantee
2. Excludes scheduled maintenance windows
3. Monthly uptime reports provided
4. Response times:
 - 4.1. Critical issues: 4 business hours
 - 4.2. High priority: 8 business hours
 - 4.3. Normal priority: 24 business hours
 - 4.4. Low priority: 48 business hours

Billing Terms

- Monthly in advance
- User count assessed at start of month
- Net-30 payment terms
- Quarterly user count reviews
- Annual contract with rolling renewal

Additional Notes

- Scheduled maintenance conducted outside business hours
- Emergency maintenance communicated with 2-hour notice
- User count based on active accounts in previous month
- Storage upgrades available on request
- Custom development charged separately

Billing & Reconciliation

Standard payment terms

- All invoices payable within 14 days
- Electronic payment preferred
- Bank details provided on invoice
- Payment reference must include invoice number

Milestone requirements

- Written approval is required for milestone completion
- 5 business days review period per milestone



- Payment due after milestone approval
- Standard revisions included
- Change requests are billed separately

Change Request Process

- Must be approved in writing
- Quoted separately from main project
- 50% deposit on approved changes
- Balance due upon completion

Document Digitisation

- A fee R72.00 will be charged for every document page that must be digitised for the application
- Payable upon the completion of Phase 2

Initial Period (First 3 Months)

During the initial three-month period, all management and support services are included in the development cost. This phase encompasses:

- Coverage of all issues related to the original specifications.
- A full system optimization and stabilization period.
- Inclusion of all hosting and infrastructure costs.

Payment Terms

- Emergency work invoiced immediately
- Standard changes billed monthly
- Contingency drawdown reported monthly
- 14-day payment terms for emergency work
- 30-day terms for standard changes

Documentation Requirements

- All changes documented in change log
- Monthly reconciliation reports
- Quarterly contingency status reports
- Annual cost analysis and recommendations



Approval Workflow

- Issue identification and documentation
- Cost estimate provided
- Approval request submitted
- Work authorization Implementation
- Review and sign-off
- Cost reconciliation



Stephan Mulder

Director

stephanm@mkat.co.za

+27 83 397 0276